

Financial Markets and Institutions

Fundamentals of Financial Institutions

Why Do Financial Institutions Exist?

Reading: Chapter 7 in Mishkin

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Overview

A closer look at why financial institutions exist and how they promote economic efficiency.

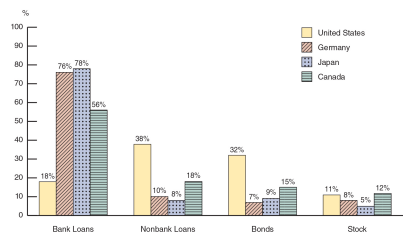
Topics include:

- Basic Facts About Financial Structure Throughout the World
- Transaction Costs
- Asymmetric Information: Adverse Selection and Moral Hazard
 - The Lemons Problem: How Adverse Selection Influences Financial Structure
 - How Moral Hazard Affects the Choice Between Debt and Equity Contracts
 - How Moral Hazard Influences Financial Structure in Debt Markets
 - Conflicts of Interest.

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Sources of External Funds for Nonfinancial Businesses

Comparison of the US, Germany, Japan, and Canada



Sources: Andreas Hackethal and Reinhard H. Schmidt, "Financing Patterns: Measurement Concepts and Empirical Results," Johann Wolfgang Goethe-Universität Working Paper No. 125, January 2004. The data are from 1970-2000 and are gross flows as percentages of the total, not including trade and other credit data, which are not available.

Although many aspects of these countries are quite different, the sources of financing are somewhat consistent, with the US being different in its focus on debt.

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Facts of Financial Structure

1. Stocks are not the most important source of external financing for businesses.
2. Issuing marketable debt and equity securities is not the primary way in which businesses finance their operations.
3. Indirect finance, which involves the activities of financial intermediaries, is many times more important than direct finance, in which businesses raise funds directly from lenders in financial markets.
4. Financial intermediaries, particularly banks, are the most important source of external funds used to finance businesses.

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Facts of Financial Structure

5. The financial system is among the most heavily regulated sectors of economy.
6. Only large, well-established corporations have easy access to securities markets to finance their activities.
7. Collateral is a prevalent feature of debt contracts for both households and businesses.
8. Debt contracts are typically extremely complicated legal documents that place substantial restrictions on the behavior of the borrowers.

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For the rest of our lecture we will look
at the reasons behind these facts

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Transaction Costs

- Transaction Costs are the time and money spent in carrying out financial transactions.
 - e.g. researching the best earning/safest stocks might be too costly if you only have only £ 100 to invest.
- Financial Intermediaries reduce the transaction costs by developing expertise and taking advantage of economies of scale
 - e.g. mutual funds
- Financial Intermediaries provide customers with **liquidity services**, services that make it easier for customers to conduct transactions
 - E.g. Banks provide depositors with bank accounts that enable them to pay their bills easily, and even earn interest.

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Asymmetric Information: Adverse Selection and Moral Hazard

- Asymmetric information: One party lacks crucial information about another party, impacting decision-making.
- We will focus on two specific forms of asymmetric information
 - Adverse selection
 - Moral hazard

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Asymmetric Information: Adverse Selection and Moral Hazard

- Adverse Selection
 1. Occurs when one party in a transaction has better information than the other party
 2. Before transaction occurs
 3. Potential borrowers most likely to produce adverse outcomes are ones most likely to seek a loan and be **selected**

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Asymmetric Information: Adverse Selection and Moral Hazard

- Moral Hazard
 1. Occurs when one party has an incentive to behave differently once an agreement is made between parties
 2. After transaction occurs
 3. Hazard that borrower has incentives to engage in undesirable (immoral) activities making it more likely they won't pay loan back

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Asymmetric Information: Adverse Selection and Moral Hazard

- The analysis of how asymmetric information problems affect behavior is known as **agency theory**.
- We will now use these ideas of adverse selection and moral hazard to explain how they influence financial structure.

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The Lemons Problem: How Adverse Selection Influences Financial Structure

- Lemons Problem in Used Cars
 1. If we can't distinguish between "good" and "bad" (lemons) used cars, we would be willing to pay only an **average** of good and bad car values
 2. Result: Good cars won't be sold, and the used car market will function inefficiently.
- What helps us avoid this problem with used cars?

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The Lemons Problem: How Adverse Selection Influences Financial Structure

- Lemons Problem in Securities Markets
 - If we can't distinguish between good and bad securities, willing to pay only **average** of good and bad securities' value.
 - Result: Good securities are undervalued and firms won't issue them; bad securities are overvalued so too many issued.

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The Lemons Problem: How Adverse Selection Influences Financial Structure

- Lemons Problem in Securities Markets

Investors don't want buy bad securities, so market don't function well

 - Explains Fact # 1 and # 2: Stocks are not the most important source of external financing for businesses. Issuing marketable debt and equity securities is not the primary way in which businesses finance their operations.
 - Also explains Fact # 6 Less asymmetric info for well-known firms, so smaller lemons problem.

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Tools to Help Solve Adverse Selection (Lemons) Problems

- Government Regulation to Increase Information (explains Fact # 5)
 - e.g. annual audits of publicly listed companies
- Financial Intermediation
 - Analogy to solution to lemons problem provided by used car dealers
- Collateral and Net Worth requirements (Explains Fact # 7)

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How Moral Hazard Affects the Choice Between Debt and Equity Contracts

- Moral Hazard in Equity Contracts: the Principal-Agent Problem
 1. Result of separation of ownership by stockholders (**principals**) from control by managers (**agents**)
 2. Managers act in own rather than stockholders' interest

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How Moral Hazard Affects the Choice Between Debt and Equity Contracts

Suppose you become a silent partner in an ice cream store, providing 90% of the equity capital (£ 9000). The other owner, Steve, provides the remaining £ 1,000 and will act as the manager. If Steve works hard, the store will make £ 50,000 after expenses, and you are entitled to £ 45,000 of it.

However, Steve doesn't really value the £ 5,000 (his part), so he goes to the beach, relaxes, and even spends some of the "profit" on art for his office. How do you, as a 90% owner, give Steve the proper incentives to work hard?

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How Moral Hazard Affects the Choice Between Debt and Equity Contracts

- Tools to Help Solve the Principal-Agent Problem
 1. Production of Information: Monitoring
 2. Government Regulation to Increase Information
 3. Financial Intermediation (e.g., venture capital)
 4. Debt Contracts
- Explains Fact # 1: Why debt is used more than equity?

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How Moral Hazard Influences Financial Structure in Debt Markets

- Even with these advantages, debt is still subject to moral hazard. In fact, debt may create an incentive to take on very risky projects.
 - Most debt contracts require the borrower to pay a fixed amount (interest) and keep any cash flow above this amount.
 - e.g what if a firm owes £ 100 in interest, but only has £ 90?
 - It is essentially bankrupt. The firm “has nothing to lose” by looking for “risky” projects to raise the needed cash.

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How Moral Hazard Influences Financial Structure in Debt Markets

- Tools to Help Solve Moral Hazard in Debt Contracts
 1. Net Worth and Collateral
 2. Monitoring and Enforcement of Restrictive Covenants.Examples are covenants that ...
 - Discourage undesirable behavior
 - Encourage desirable behavior
 - Keep collateral valuable
 - Provide information

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How Moral Hazard Influences Financial Structure in Debt Markets

- Tools to Help Solve Moral Hazard in Debt Contracts
 3. Financial Intermediation—banks and other intermediaries have special advantages in monitoring
- Explains Facts # 1–4

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Conflicts of Interest

- Conflicts of interest are a type of moral hazard that occurs when a person or institution has multiple interests, and serving one interest is detrimental to the other.
- Three classic type of conflicts developed in financial institutions are:
 1. Auditing and Consulting in Accounting Firms
 2. Underwriting and Research in Investment Banking
 3. Credit Assessment and Consulting in Rating Agencies

We will learn more about the last two once in weeks 6 and 7 of semester 1.
